



OPEN Microbiological profile and antimicrobial resistance in diabetic foot infections: a cross-sectional study from a low- to middle-income country

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Diabetic foot infections (DFIs) are serious complications of diabetes mellitus, yet little is known about the bacterial profile and antibiotic resistance patterns among patients with diabetic foot ulcers in this region. This study aimed to assess the prevalence and antimicrobial resistance of DFIs in the main referral governmental hospitals in the northern West Bank, Palestine. This descriptive, retrospective cross-sectional study was conducted in the Vascular Surgery Department at Rafidia Surgical Hospital in Nablus, Palestine. The medical records of patients with DFIs in 2022 were reviewed, and the data were analysed via IBM SPSS for Windows v.21.0. Among the 118 patients, 58.5% were male, and 98.3% had type II diabetes mellitus. Gram-negative bacteria were more prevalent, accounting for 58.8% of the cases, whereas gram-positive bacteria accounted for 41.2%. *Staphylococcus aureus* (25.4%), *Escherichia coli* (14.0%), and *Pseudomonas aeruginosa* (11.4%) were the most frequently isolated pathogens. Extended-spectrum β -lactamase (ESBL)-producing strains were detected among *E. coli* and *Klebsiella pneumoniae* isolates, and carbapenemase-producing Enterobacterales (CPEs) were also identified. Among gram-positive bacteria, *Staphylococcus aureus* is methicillin-resistant, and *Enterococcus* species are vancomycin-resistant. Multidrug-resistant (MDR) isolates, as *Acinetobacter baumannii*, are present in both gram-positive and gram-negative groups. A longer infection duration and longer hospital stay were significantly associated with increased amputation risk ($p = 0.044$ and $p < 0.001$, respectively), highlighting the need of early intervention and targeted antibiotic therapy. This study revealed a high prevalence of multidrug-resistant pathogens among diabetic foot infections in northern Palestine, with *Staphylococcus aureus*, *Escherichia coli*, and *Pseudomonas aeruginosa* as the leading causative organisms. The notable presence of methicillin-resistant *S. aureus* and extended-spectrum β -lactamase-producing Enterobacterales underscores the urgent need for robust antimicrobial stewardship and regular local surveillance to guide empirical therapy. Early diagnosis, appropriate antibiotic selection, and timely intervention are crucial to improving clinical outcomes and reducing amputation risk in this vulnerable population.

Keywords Diabetic foot infections, Antimicrobial resistance, Gram-negative bacteria, Methicillin-resistant *Staphylococcus aureus*, Extended-spectrum beta-lactamase, Multidrug resistance

Abbreviations

BMI Body mass index

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CLSI	Clinical and Laboratory Standards Institute
CPE	Carbapenemase-producing enterobacterales
DFIs	Diabetic foot infections
DFUs	Diabetic foot ulcers
ESBL	Extended-spectrum beta-lactamase
HDL	High-density lipoprotein
IRB	Institutional Review Board
MDR	Multidrug-resistant
MRSA	Methicillin-resistant <i>Staphylococcus aureus</i>
SPSS	Statistical Package for Social Sciences software

Diabetes mellitus is a widespread noncommunicable disease affecting approximately 9.3% of the global population and more than 463 million individuals worldwide¹. It is associated with numerous complications, including retinopathy, nephropathy, cardiovascular diseases, and foot injuries, which significantly affect patients' quality of life^{2,3}. Among these complications, diabetic foot ulcers (DFUs) pose a significant burden, leading to high morbidity and mortality rates⁴. The global prevalence of DFUs is approximately 6.3%, which is influenced by sociodemographic and self-care factors^{5,6}. In Palestine, DFUs have been linked to sensory loss, smoking, neuropathy, and inadequate self-care, highlighting the need for better prevention and management⁷.

Diabetic foot infections (DFIs) are a leading cause of hospitalization and amputation in diabetic patients⁸. They are often polymicrobial, commonly involving *Staphylococcus aureus*, *Escherichia coli*, *Klebsiella pneumoniae*, and *Proteus* spp., along with anaerobes⁹. Diabetic foot osteomyelitis, a severe form of infection affecting up to 68% of hospitalized DFI patients, further complicates management and increases healthcare costs^{10,11}. Risk factors include chronic ulcers, neuropathy, renal disease, and peripheral arterial disease^{12,13}. Management requires microbiological assessment, targeted antibiotics, wound care, and sometimes surgery^{8,14}. The rise of multidrug-resistant (MDR) organisms underscores the importance of antimicrobial stewardship¹⁵.

Globally, DFIs affect 18.6 million individuals annually, with ulceration preceding nearly 80% of amputations¹⁶. MDR pathogens such as *Pseudomonas aeruginosa* and ESBL-producing bacteria complicate therapy¹⁷. Regional studies, including those from Palestine, Egypt, Lebanon, and India, have shown variation in microbial profiles and resistance patterns^{18–21}. Mortality risk factors include peripheral artery disease, polymicrobial infections, and high thrombocyte levels^{22–24}.

With a five-year mortality rate exceeding 50%, primarily due to cardiovascular disease and infections, early intervention integrating infection control, ischemia management, and comorbidity treatment is crucial for reducing adverse outcomes²⁵.

In low- and middle-income countries such as Palestine, the burden of diabetes mellitus and its complications is increasing, but data on DFI microbiology and resistance are limited. Understanding the local microbial landscape is essential for guiding empirical therapy and improving outcomes. This study aimed to assess the prevalence and antimicrobial resistance patterns of DFIs in patients admitted to referral hospitals in the northern West Bank, Palestine. The findings provide evidence to guide management strategies and inform antimicrobial stewardship in this resource-limited setting.

Methods

Study design and population

The study was conducted in a retrospective cross-sectional design at Rafidia Surgical Hospital, which is one of the main referral governmental hospitals north of the West Bank, Palestine. This study included diabetic patients admitted to Rafidia Surgical Hospital with foot infections. The medical records of patients admitted between January 2022 and December 2022 were reviewed.

Sample size and calculation

The inclusion criteria were adult patients (18 years or older) with a confirmed diagnosis of type 1 or type 2 diabetes mellitus and a clinically diagnosed foot infection. The exclusion criteria included patients with osteomyelitis, severe peripheral arterial disease, systemic infections unrelated to diabetic foot infections, or incomplete medical records. Patients who sought care at Rafidia Surgical Hospital and met the inclusion criteria were included. A total of 118 participants were recruited and included via a convenience sampling method.

Study tools and data collection

Data were collected from the medical records of the patients as well as microbiology laboratory reports. A data collection form was developed specifically for this study.

The collected data included the sociodemographic and clinical characteristics of the patients, including age, sex, marital status, education, occupation, place of residence, smoking status, weight, height, and monthly income. According to the Palestinian Central Bureau of Statistics, a monthly income of less than 3500 NIS was categorized as low, whereas an income of 3500 NIS or more was considered high²⁶.

Clinical data, including the type of diabetes mellitus, antidiabetic medications, duration of diabetes mellitus, ulcer history (defined as the time elapsed between the onset of symptoms and hospital admission), previous hospitalizations, and prior use of antimicrobial agents, were also recorded. Data on prior antibiotic use within 90 days were collected; however, prior microbiological culture results were not consistently available in the medical records and were therefore not analysed. Comorbidities such as hypertension, coronary heart disease, stroke, peripheral arterial disease, chronic kidney disease, and hemodialysis were documented.

Finally, patient outcomes, including whether the ulcer healed or remained unhealed, whether amputation was performed, whether the patient survived or died, and the occurrence of diabetic complications such as nephropathy, retinopathy, peripheral arterial disease, and autonomic neuropathy, were recorded.

Microbiological identification, susceptibility testing, and resistance classification

The VITEK[®] 2 Compact automated microbiology system (bioMérieux, Marcy-l'Étoile, France) was employed for microorganism identification and antibiotic susceptibility testing. It has also been utilized for detecting extended-spectrum beta-lactamase (ESBL) producers and carbapenemase-producing *Enterobacterales* (CPEs). Methicillin resistance was assessed via the use of ceftiofur. Resistance rates were determined following the guidelines of the Clinical and Laboratory Standards Institute (CLSI). MDR was defined as acquired nonsusceptibility to at least one agent in three or more antimicrobial categories²⁷.

Statistical analysis

Data were analyzed using IBM SPSS version 21.0 (IBM Corp., Armonk, NY, USA). The data were tested for normality of distribution via the Kolmogorov–Smirnov or Shapiro–Wilk test. Because the data were not normally distributed, as confirmed by the Kolmogorov–Smirnov and Shapiro–Wilk tests, nonparametric statistical analyses were applied. Continuous variables were summarized using medians and interquartile ranges (IQRs), and categorical variables were presented as frequencies and percentages. Comparisons between two independent groups (e.g., patients with and without major amputation) were performed using the Mann–Whitney U test for continuous variables and the Chi-square test or Fisher's exact test for categorical variables, as appropriate. A p value of <0.05 was considered statistically significant.

Ethical considerations

The study protocol was approved by the Institutional Review Board (IRB) of An-Najah National University [IRB Reference #: Mas. Feb,2023/19], and the necessary approval was obtained from the Ministry of Health and Rafidia Surgical Hospital. The research adhered to the principles of the Declaration of Helsinki. The participants were informed about the purpose of the study, and oral consent was obtained before their participation. Confidentiality and privacy were maintained throughout the study, with access to patient data restricted to the research team.

Results

Sociodemographic and clinical characteristics of the patients

Among the 118 patients, the study sample consisted primarily of male participants (58.5%), and the majority were married (73.7%). Over half of the participants were smokers (63.6%), and most had completed only school-level education (77.1%). The majority of the participants had a low income, earning less than 3500 NISs (82.2%). With respect to place of residence, half of the participants lived in villages (50.0%), whereas a smaller portion resided in cities (33.1%). A notable proportion of the participants were classified as obese (61.0%), and nearly 98.3% of them had type II diabetes mellitus. The most commonly used antidiabetic medication was insulin (62.7%), followed by sitagliptin (17.8%) and metformin (14.4%). Other details can be found in Table 1.

Clinical characteristics and laboratory test results of the participants

The most common reasons for hospitalization among the participants were heel pressure ulcers (44.9%) and infected toes (39.0%). Half of the patients had diabetic nephropathy (50.0%), 29.7% had retinopathy, and 11.0% had neuropathy. The most prevalent comorbidities were hypertension (65.3%) and hyperlipidemia (55.1%), with a significant number also having heart failure (22.9%) and respiratory disease (33.1%). Approximately half of the patients had been hospitalized in the last 90 days (47.5%), and a large majority had used antibiotics during this period (88.1%). Regarding lipid profiles, 43.2% had abnormal total cholesterol levels, and 47.5% had elevated triglycerides. Additionally, 89.8% of patients had an HbA1c level of 7% or higher, indicating poor glycemic control. These details are shown in Table 2.

Frequency of isolated microorganisms

Table 3 shows the distribution of the isolated bacteria, where gram-negative bacteria were more predominant, accounting for 58.8% of the total isolates, and gram-positive bacteria accounted for 41.2% of the isolates. Among the gram-negative isolates, *Escherichia coli* was the most prevalent (14.0%), followed by *Pseudomonas aeruginosa* (11.4%) and *Proteus mirabilis* (8.8%). Among the gram-positive isolates, *Staphylococcus aureus* accounted for 25.4%, whereas coagulase-negative staphylococci and *Enterococcus* spp. accounted for 6.1% and 5.3%, respectively. Notably, 4.4% of the cultures showed no growth.

Antibiotic susceptibility of the most frequent gram-positive bacteria

Our study showed that *Staphylococcus aureus* was highly susceptible to vancomycin (93.1%), gentamicin (89.7%), tetracycline (72.4%), and trimethoprim/sulfamethoxazole (75.9%), susceptible to clindamycin (48.3%) and ciprofloxacin (41.4%), and had low susceptibility to ampicillin (6.9%) and cefotaxime (24.1%). Coagulase-negative staphylococci were completely susceptible to vancomycin and susceptible to both tetracycline and doxycycline (both 85.7%) but were moderately susceptible to both gentamicin (57.1%) and trimethoprim/sulfamethoxazole (42.9%) and were less susceptible to erythromycin (14.3%) and cefotaxime (14.3%). *Enterococcus* spp. were susceptible to vancomycin (83.3%), ampicillin (83.3%), and gentamicin (66.7%) and were moderately susceptible to ciprofloxacin (50%) and piperacillin/tazobactam (33.3%) (Table 4).

Characteristics	Categories	N N = 118	%
Sex	Male	69	58.5
	Female	49	41.5
Marital status	Single	5	4.2
	Married	87	73.7
	Widowed	26	22.0
Smoking status	No	43	36.4
	Yes	75	63.6
Educational level	School	91	77.1
	University	27	22.9
Income	< 3500 NIS	97	82.2
	≥ 3500 NIS	21	17.8
Place of residence	Village	59	50.0
	City	39	33.1
	Refugee camp	20	16.9
Body mass index (BMI)	Normal weight (< 24.9 kg/m ²)	14	11.9
	Overweight (25–29.9 kg/m ²)	32	27.1
	Obese (≥ 30 kg/m ²)	72	61.0
Diabetes type	Type I	2	1.7
	Type II	116	98.3
Antidiabetic medications	Insulin	74	62.7
	Metformin	17	14.4
	Vildagliptin/Metformin	5	4.2
	Sitagliptin	21	17.8
	Glimepiride	1	0.8

Table 1. Sociodemographic and clinical characteristics of the patients.

Antibiotic susceptibility of the most common gram-negative bacteria

Escherichia coli exhibited high susceptibility to meropenem (100%), amikacin (93.8%), and piperacillin/tazobactam (81.3%); moderate susceptibility to imipenem (68.8%) and gentamicin (50%); and low susceptibility to ampicillin (6.3%) and cefotaxime (25%). *Pseudomonas aeruginosa* was susceptible to ceftazidime (84.6%), meropenem (76.9%), piperacillin/tazobactam (76.9%), gentamicin (69.2%), and cefepime (69.2%). *Proteus mirabilis* was highly sensitive to piperacillin/tazobactam and meropenem (100% each), amikacin (90%), and cefotaxime (90%), moderately sensitive to cefepime and gentamicin (70% each), and relatively less sensitive to ciprofloxacin (50%) and levofloxacin (10%). *Klebsiella pneumoniae* had good sensitivity to meropenem and amikacin (87.5% each) but had low sensitivity to ceftazidime, cefotaxime, and ceftriaxone (12.5% each), with intermediate sensitivity to imipenem and ciprofloxacin (37.5% each). The most prevalent organisms and their antibiotic sensitivities are provided in Table 4.

Frequency of multidrug-resistant organisms

MDR organisms were prevalent, accounting for 32.6% of all isolates, and *Staphylococcus aureus* had the highest proportion of MDR at 48.3%, followed by *Klebsiella pneumoniae* (37.5%) and *Escherichia coli* (25.0%). Methicillin-resistant *Staphylococcus aureus* (MRSA) was also prevalent, accounting for 31.0% of the *S. aureus* isolates. Resistance against carbapenem strains was also observed in 15.4% of *Pseudomonas aeruginosa* strains and 12.5% of *Klebsiella pneumoniae* strains. Although vancomycin resistance was indeed low in all the strains (3.4%), it was observed in 6.9% of the *S. aureus* strains and 16.7% of the *Enterococcus* spp. as demonstrated in Table 5.

Associations between duration of infection and length of hospital stay with amputation

The patients who had major amputations had increased infection rates and hospital stay lengths, as shown in Table 6. Patients who underwent major amputation had a much greater duration of infection, with a median of 2.7 months, than patients who did not undergo major amputation, with a median of 2.0 months. In addition, the length of hospital stay was much longer among those who underwent major amputation, with a median of 22.0 days, than among those without major amputation, with a median of 11.0 days. These differences were statistically significant and indicated that there was a strong relationship between prolonged infection, longer hospital stays, and the likelihood of amputation.

Discussion

DFIs are a major cause of morbidity and mortality globally, resulting in compromised quality of life for infected individuals²⁸. The present study focused on the microbiological pattern and antimicrobial resistance of DFIs in

Characteristics	Categories	n	%
Reason for hospitalization	Heel pressure ulcers	53	44.9
	Infected toe	46	39.0
	Plantar abscess	14	11.9
	Amputated stump infection	5	4.2
Diabetes complications	Nephropathy	59	50.0
	Retinopathy	35	29.7
	Neuropathy	13	11.0
Comorbidities and history of hospitalization	End-stage renal disease	14	11.9
	Heart failure	27	22.9
	Hypertension	77	65.3
	Hyperlipidemia	65	55.1
	Chronic kidney disease	12	10.2
	Chronic liver disease	5	4.2
	Neurological disease	19	16.1
	Respiratory disease	39	33.1
	Malignancy	22	18.6
History of hospitalization and use of antibiotics	Previous hospitalization in the last 90 days	56	47.5
	Previous ICU admission in the last 90 days	1	0.8
	Use of antibiotics in the last 90 days	104	88.1
Total cholesterol	Normal (< 200 mg/dL)	67	56.8
	Abnormal (≥ 200 mg/dL)	51	43.2
Triglyceride	Normal (< 150 mg/dL)	62	52.5
	Abnormal (≥ 150 mg/dL)	56	47.5
Low density lipoprotein cholesterol (LDL-C)	Normal (< 130 mg/dL)	106	89.8
	Abnormal (≥ 130 mg/dL)	12	10.2
High density lipoprotein cholesterol (HDL-C)	Normal (≥ 40 mg/dL)	55	46.6
	Abnormal (< 40 mg/dL)	63	53.4
HbA _{1c}	Normal (< 7%)	12	10.2
	Abnormal (≥ 7%)	106	89.8

Table 2. Clinical characteristics and laboratory test results of the participants. ICU: intensive care unit.

Palestine, addressing a gap in the literature. The study population included 118 patients, which is consistent with the findings of previous studies²⁹. The median age was 62 years, and the median BMI was 31.3 kg/m², both of which are known risk factors for type II diabetes mellitus³⁰. More than half of the patients were male, which is consistent with global trends in diabetes mellitus incidence³¹, and some had other risk factors, such as smoking³² and low income, which have been linked to worse outcomes in patients with diabetes mellitus³³. The majority of patients also had high HbA_{1c} levels and poor cholesterol profiles, again indicating poor control of diabetes mellitus. Some patients also have advanced disease, often requiring insulin therapy, highlighting the need for healthcare professionals to treat complications such as DFIs.

The participants in this study were admitted to the hospital due to infections such as heel pressure ulcers and infected toes, among other diabetes-related complications such as nephropathy and retinopathy¹⁰. The majority of these patients have comorbidities such as hypertension and hyperlipidaemia, which are major risk factors for diabetic foot infection complications³⁴. Several patients had been recently treated with antibiotics, which could have affected the way their infections were being treated. Gram-positive and gram-negative bacteria, e.g., *Staphylococcus aureus*, *Escherichia coli*, *Pseudomonas aeruginosa*, and *Proteus mirabilis*, were the most common pathogens isolated, which was in accordance with the results of a meta-analysis conducted in China³⁵.

Importantly, the elevated sensitivity (~ 90%) of *P. mirabilis* to ceftazidime and cefotaxime observed in this study may reflect partial susceptibility within the local strain population. This is likely due to the heterogeneous expression of extended-spectrum β-lactamases (ESBLs) and the relatively low prevalence of AmpC β-lactamase activity. In contrast, the reduced susceptibility to cefepime (~ 70%) may be attributed to the presence of inducible or derepressed AmpC β-lactamases and other resistance mechanisms that compromise the activity of this fourth-generation cephalosporin. These findings emphasize the multifactorial nature of *Proteus mirabilis* β-lactam resistance and highlight the importance of local antimicrobial susceptibility testing in guiding empirical antibiotic therapy decisions³⁶.

Most of these pathogens are multidrug resistant, as reported in other studies^{37–39}, which implies treatment challenges. These findings underscore the need for recent antibiotic resistance findings to inform local treatment guidelines optimally⁴. These findings highlight the urgent need to reinforce antimicrobial stewardship policies in Palestinian healthcare settings, including guidelines for empirical antibiotic use and restrictions on over-the-counter antibiotic access.

Culture result	n	%
No growth	4	
Gram-positive bacteria	47	41.2
<i>Staphylococcus aureus</i>	29	25.4
<i>Coagulase-negative staphylococci</i>	7	6.1
<i>Enterococcus</i> spp.	6	5.3
<i>Streptococcus agalactiae</i>	3	2.6
<i>Viridans streptococci</i>	1	0.9
<i>Bacillus</i> spp.	1	0.9
Gram-negative bacteria	67	58.8
<i>Escherichia coli</i>	16	14.0
<i>Pseudomonas aeruginosa</i>	13	11.4
<i>Proteus mirabilis</i>	10	8.8
<i>Klebsiella pneumoniae</i>	8	7.0
<i>Morganella morganii</i>	5	4.4
<i>Acinetobacter baumannii</i>	4	3.5
<i>Citrobacter</i> spp.	2	1.8
<i>Enterobacter cloacae</i>	2	1.8
<i>Klebsiella oxytoca</i>	2	1.8
<i>Serratia marcescens</i>	2	1.8
<i>Proteus vulgaris</i>	1	0.9
<i>Providencia stuartii</i>	1	0.9
<i>Acinetobacter</i> spp.	1	0.9

Table 3. Frequencies of isolated microorganisms.

Antibiotic	Gram-positive						Gram-negative							
	<i>Staphylococcus aureus</i> (n = 29)		<i>Coagulase-negative staphylococci</i> (n = 7)		<i>Enterococcus</i> spp. (n = 6)		<i>Escherichia coli</i> (n = 16)		<i>Pseudomonas aeruginosa</i> (n = 13)		<i>Proteus mirabilis</i> (n = 10)		<i>Klebsiella pneumoniae</i> (n = 8)	
	n	%	n	%	n	%	n	%	n	%	n	%	n	%
Ampicillin	2	6.9	0	0.0	5	83.3	1	6.3	-	-	3	30.0	0	0.0
Amoxicillin/clavulanic acid	6	20.7	ND	ND	1	16.7	10	62.5	-	-	8	80.0	6	75.0
Piperacillin/tazobactam	1	3.4	ND	ND	2	33.3	13	81.3	10	76.9	10	100.0	6	75.0
Ceftazidime	ND	ND	ND	ND	0	0.0	3	18.8	11	84.6	9	90.0	1	12.5
Cefotaxime	7	24.1	1	14.3	0	0.0	4	25.0	1	7.7	9	90.0	1	12.5
Ceftriaxone	1	3.4	ND	ND	0	0.0	4	25.0	1	7.7	8	80.0	1	12.5
Cefepime	0	0.0	ND	ND	0	0.0	4	25.0	9	69.2	7	70.0	3	37.5
Meropenem	ND	ND	ND	ND	ND	ND	16	100.0	10	76.9	10	100.0	7	87.5
Imipenem	ND	ND	ND	ND	ND	ND	11	68.8	9	69.2	3	30.0	3	37.5
Vancomycin	27	93.1	7	100.0	5	83.3	1	6.3	ND	ND	ND	ND	0	0.0
Erythromycin	9	31.0	1	14.3	1	16.7	ND	ND	ND	ND	ND	ND	ND	ND
Clindamycin	14	48.3	2	28.6	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
Tetracycline	21	72.4	6	85.7	0	0.0	0	0.0	ND	ND	ND	ND	ND	ND
Doxycycline	16	55.2	6	85.7	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
Gentamicin	26	89.7	4	57.1	4	66.7	8	50.0	9	69.2	7	70.0	6	75.0
Amikacin	4	13.8	ND	ND	1	16.7	15	93.8	9	69.2	9	90.0	7	87.5
Ciprofloxacin	12	41.4	2	28.6	3	50.0	4	25.0	5	38.5	5	50.0	3	37.5
Levofloxacin	0	0.0	ND	ND	0	0.0	1	6.3	4	30.8	1	10.0	ND	ND
Trimethoprim/sulfamethoxazole	22	75.9	3	42.9	0	0.0	4	25.0	1	7.7	4	40.0	4	50.0

Table 4. The antibiotic susceptibility of the most frequent organisms. ND: Not determined.

Organism	Total Isolates	ESBL	CPE	Vancomycin-resistant	Methicillin-resistant	MDR
Gram-positive						
<i>Staphylococcus aureus</i>	29	-	-	2 (6.9%)	9 (31.0%)	14 (48.3%)
<i>Coagulase-negative Staphylococci</i>	7	-	-	0 (0.0%)	-	2 (28.6%)
<i>Enterococcus</i> spp.	6	-	-	1 (16.7%)	-	1 (16.7%)
Gram-negative						
<i>Escherichia coli</i>	16	4 (25.0%)	0 (0.0%)	-	-	4 (25.0%)
<i>Pseudomonas aeruginosa</i>	13	-	2 (15.4%)	-	-	2 (15.4%)
<i>Proteus mirabilis</i>	10	-	0 (0.0%)	-	-	3 (30.0%)
<i>Klebsiella pneumoniae</i>	8	2 (25.0%)	1 (12.5%)	-	-	3 (37.5%)
Total	89	6 (6.7%)	4 (4.5%)	3 (3.4%)	9 (10.1%)	29 (32.6%)

Table 5. Frequencies of multidrug-resistant (MDR) organisms. ESBL: extended-spectrum β -lactamase, CPE: carbapenemase-producing *Enterobacterales*, MDR: multidrug resistant, MRSA: methicillin-resistant *Staphylococcus aureus*.

Variable	Major amputation	Q1	Median	Q3	<i>p</i>
Duration of infection (month)	No	1.0	2.0	3.5	0.044
	Yes	2.0	2.7	6.0	
Length of hospital stay (day)	No	7.5	11.0	20.0	<0.001
	Yes	20.0	22.0	34.0	

Table 6. Association between duration of infection and length of hospital stay with amputation.

The study also confirmed that some patients presented with severe lower limb complication, that required amputation⁴⁰. The intensity of DFIs makes prevention early enough before it is difficult to treat. Ulcer size was related to other diabetes mellitus complications, comorbidities, hospitalizations, and major amputations, which is in agreement with previous findings⁴¹. These findings reaffirm that better management of diabetes mellitus and comorbid conditions can avoid DFIs and contribute to better patient outcomes. Although we observed notable antimicrobial resistance patterns, we were unable to correlate specific pathogens or resistance phenotypes with outcomes such as amputation or mortality because of the limited subgroup analysis. Future prospective studies should explore these associations.

Several limitations need to be considered when the findings of the study are interpreted. The retrospective design employs available medical records, which may be incomplete or inexact, hence potentially introducing bias. As this was a single-center study conducted at Rafidia Surgical Hospital, generalizability is uncertain in other regions or contexts of healthcare provision. A cross-sectional design does not allow for the establishment of changes in bacterial patterns or resistance trends over time. Additionally, the use of convenience sampling may have introduced selection bias, as the study population may not fully represent all DFI patients in the region. Moreover, the absence of stratified analysis by variables such as age, ulcer severity, and comorbidities may limit our ability to assess their impact on infection patterns and outcomes. Data on prior antibiotic use were collected, but prior microbiological culture results were not consistently available, which may have limited the assessment of previous infection profiles and antibiotic resistance patterns. This study also provides a limited correlation analysis of the relationships among some pathogens, resistance patterns, and clinical outcomes, such as amputation rates or mortality. Additionally, the study did not employ multivariate regression or calculate odds ratios, which could have enhanced the analysis by adjusting for confounding variables. Finally, a small number of patients with prior amputations (5, 4.2%) who were admitted with stump infections were included in the study. This may have introduced bias, as post-amputation anatomical and biomechanical changes could influence infection recurrence, bacterial colonization, and treatment outcomes.

Conclusions

This study provides significant insight into the microbiological trends and antimicrobial resistance of diabetic foot infection in the northern West Bank, Palestine. The findings are indicative of an intense presence of gram-negative organisms, such as *Escherichia coli* and *Klebsiella pneumoniae*, and the intensity of *Staphylococcus aureus* among the gram-positive isolates. This study highlights the disconcerting prevalence of antimicrobial resistance, such as elevated rates of ESBL production among gram-negative bacteria and MRSA production in *S. aureus*. The dominance of multidrug-resistant organisms, especially *Acinetobacter baumannii*, also makes the treatment of DFIs challenging in this region. This study highlights the urgent need for implementing hospital-based antimicrobial stewardship programs, establishing regional resistance surveillance, and promoting early multidisciplinary management to prevent severe complications such as amputation.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author upon request.

Received: 27 March 2025; Accepted: 16 December 2025

Published online: 19 December 2025

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Acknowledgements

We would like to acknowledge the support of An-Najah National University and its faculty for providing the necessary resources and a conducive environment for research.

Author contributions

S.Z. and A.A.T. contributed to the idea conception. S.Z. and A.A.T. planned the study and its methodology and critically reviewed and finalized the manuscript. A.A. collected the data. A.A. and B.A. performed the data analysis, and B.A., A.A. and D.A. prepared the initial draft of the manuscript under A.A.T. supervision. All authors revised and approved the submission of the final version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Ethical approval and consent to participate

Verbal informed consent was obtained from all participants prior to their inclusion in the study. The study protocol received ethical approval from the Institutional Review Board (IRB) of An-Najah National University (IRB Reference #: Mas.Feb,2023/19), and the necessary permissions were granted by the relevant local authorities. The interviewer obtained verbal consent from each participant who agreed to take part in the study after explaining its purpose and procedures. All collected data were used solely for research purposes and were treated with strict confidentiality. As no personal identifiers were collected and the study posed minimal risk to participants, the IRB approved the use of verbal rather than written consent. All procedures were conducted in accordance with institutional guidelines, relevant national and international regulations, and the ethical principles outlined in the Declaration of Helsinki.

Additional information

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